PADERBORN UNIVERSITY

W4479 – Econometrics, winter semester 2024 / 2025

Applied Econometrics Project With R

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Problem No.	1	2	3	4	Total
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Submitted by Group No. 45:

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Problem 1

a)

Below are the results and keys findings:

Number of Observations	Error Function	Mean Intercept	Variance Intercept	Mean Slope	Variance Slope
35	Normal	3.46	4.93	0.65	0.000222
35	Chi-squared	4.04	5.17	0.643	0.000218
120	Normal	3.57	1.1	0.649	0.0000501
120	Chi-squared	3.88	1.22	0.644	0.0000524
500	Normal	3.54	0.152	0.649	0.00000706
500	Chi-squared	3.9	0.17	0.644	0.00000706

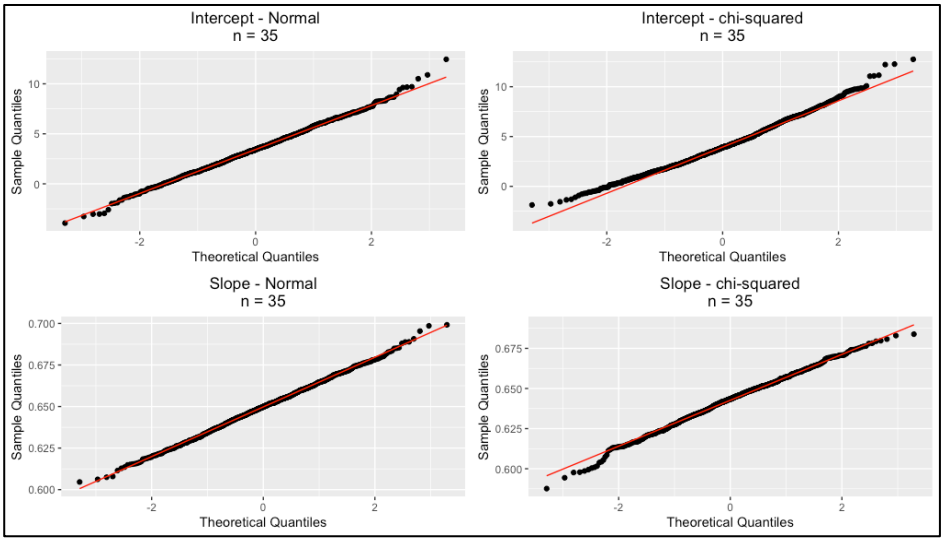
Comparisons to true values: The true intercept is β_1 =3.4. On average, the intercept estimates around 3.4-4.0. For Chi-squared errors, the mean is always higher than 3.4. Overall, the slopes estimates are close to 0.64-0.65, suggesting minimal bias from β_2 =0.65.

Effect of the sample size on variance: As (n) increases from 35 to 500, the variances decrease of both intercept and slope, reflecting the usual consistency and improved precision of OLS for larger samples.

Normal vs Chi-squared errors: The OLS provide (approximately) unbiased estimates of both β1 and β2. The sampling variance shrinks as (n) grows, and the impact of Chi-squared errors on the estimates means mostly appears in smaller sample, where skewness and heavy tails can shift estimates.

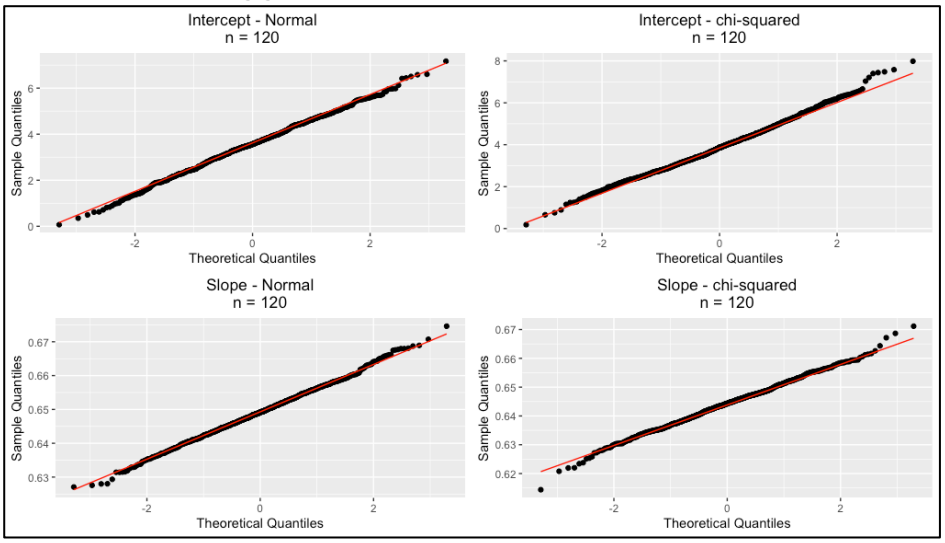
b)

For each scenario, Q–Q plots of the intercept and slope estimates were generated. A 4×4 grid was produced to display the various combinations slope, intercept and error distribution. The figure shows the resulting graphs for the case where sample Size n=35.



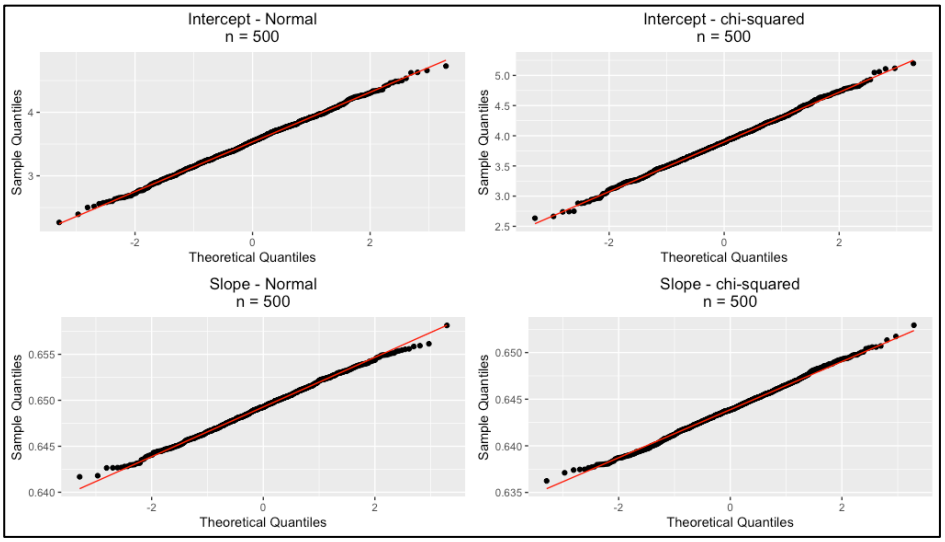
1. **Intercept- Normal:** The points closely follow the diagonal reference line, with slight deviations at the trails. Which indicates the graph Supports approximate normality.
2. **Intercept – Chi-squared:** The points deviate significantly, especially in the upper quantities, from the diagonal reference line. This indicates the graph does not support normality.
3. **Slope – Normal:** The points align well with the diagonal reference line, with minimal deviations at the tails. The Graph Support approximate normality.
4. **Slope – Chi- Squared:** Noticeable deviations, particularly in the lower quantiles, suggest a lack of fit. This indicates the graph does not support normality.

For sample size n=120 following graphs were obtained:



1. **Intercept- Normal:** The points follow the diagonal line more closely than in n= 35, with minor tail deviations, which indicates the graph supports approximate normality.
2. **Intercept – Chi- squared:** The points deviate from the reference line, although they are less pronounced compared to n= 35. The graph still does not support normality.
3. **Slope – Normal:** The alignment with the diagonal is strong, showing better conformity to normality compared to n=35. The graph supports approximate normality.
4. **Slope – Chi- Squared:** While deviations are reduced compared to n= 35, the points still diverge from the diagonal line. The graph does not support normality.

For sample size n=500 following graphs were obtained:



1. **Intercept- Normal:** The points align closely with the 45 degrees redline. The graph supports approximate normality.
2. **Intercept – Chi- squared:** The points deviate the reference line, although are less pronounced compared to n=120. The graph does not support normality.
3. **Slope – Normal:** The points align well with the red line, showing minimal deviation. This indicates that the data follows approximate normality distribution.
4. **Slope – Chi- Squared:** The points deviate from the line, particularly at the tails. This indicates the data does not follow a normal distribution.

The QQ plots (plots on the left) provide approximate normality for the slope and intercept under the normal error distribution. However, because of the obvious deviations from the line, which are consistent with the non-normal skewness of the Chi-Squared distribution, the data does not near normality under the Chi-Squared error distribution (plots on the right).

C)

The following are the findings of the Jarque-Bera test for normality for each combination, the table includes the following columns:

- JB_stat: Jarque-Bera test statistic.
- p_value: The probability of observing the data under the null hypothesis of normality.
- crit_value: Critical value at 5% significance level (5.9914 for 2 degrees of freedom).
- reject: Whether the null hypothesis of normality is rejected (TRUE if p-value < 0.05).

n	error_dist	parameter	JB_stat	p_value	crit_value	reject
35	Normal	Intercept	2.9825	0.2250896	5.9914	FALSE
35	Normal	Slope	0.0219	0.9890850	5.9914	FALSE
35	chi-squared	Intercept	26.8572	0.0000015	5.9914	TRUE
35	chi-squared	Slope	7.9158	0.0191025	5.9914	TRUE
120	Normal	Intercept	1.8626	0.3940301	5.9914	FALSE
120	Normal	Slope	1.2299	0.5406456	5.9914	FALSE
120	chi-squared	Intercept	12.1295	0.0023232	5.9914	TRUE
120	chi-squared	Slope	5.3130	0.0701922	5.9914	FALSE
500	Normal	Intercept	0.3719	0.8302878	5.9914	FALSE
500	Normal	Slope	3.1421	0.2078207	5.9914	FALSE
500	chi-squared	Intercept	0.2041	0.9029457	5.9914	FALSE
500	chi-squared	Slope	0.9097	0.6345291	5.9914	FALSE

When errors are normally distributed, intercept and slope estimates remain normal across all sample sizes, as confirmed by Jarque-Bera (JB) tests failing to reject normality. However, under chi-squared errors, small samples (n = 35) produce non-normal estimates for both parameters, while moderate samples (n = 120) show the intercept remains non-normal but the slope becomes borderline normal ($p \approx 0.07$). At n = 500, the Central Limit Theorem (CLT) dominates: estimates for both parameters approximate normality, even with non-normal errors. This demonstrates that larger samples mitigate non-normality in estimates, with the intercept being more sensitive to small-sample deviations than the slope.

D)

Using the results from question a), b), and c):

1. **Small Sample (n=35)**
   - **Normal Errors**: Slope and intercept both seem compatible with normality, passing the Jarque-Bera test.
   - **Chi-squared Errors**: Both the intercept and the slope are rejected as normal. This is consistent with the Q-Q plots in part (b), where chi-squared errors resulted in obvious tail/skew problems at small n.
2. **Moderate Sample (n=120)**
   - **Normal Errors**: Intercept and slope still look normal (no rejection).
   - **Chi-squared Errors**: Although the slope is borderline ($p \sim 0.07$) but not rejected at 5%, the intercept is rejected, indicating non-normality. The slope distribution is therefore "less obviously" non-normal, maybe because of the slope's distinct dependence on the errors and X-values.
3. **Large Sample (n=500)**
   - In large samples, the JB test does not reject either the intercept or the slope for both Normal and Chi-squared errors, suggesting approaching normality. Additionally, near-straight-line patterns are confirmed by the Q-Q plots from part (b).

Under econometric theory ordinary least squares (OLS) estimators are consistent and asymptotically normal under standard assumptions such as zero-mean and finite-variance errors. However, with small sample sizes, such as *n=35*, the presence of non-normal errors, particularly heavily skewed or

heavy-tailed distributions (e.g., chi-squared), can lead to significant deviations from normality, often resulting in rejections of normality for chi-squared distributed errors. As the sample size grows, the Central Limit Theorem ensures that the distribution of OLS estimators converges toward normality, even when the underlying errors are not normally distributed. For instance, with *n=500*, normality rejections become rare, aligning with theoretical expectations.

Hence, **all results from (a), (b), and (c)** support standard econometric predictions:

1. **At small samples**, normal errors yield intercepts and slopes that appear near-normal, while chi-squared errors can introduce skewness, causing clear deviations.
2. **At large samples**, regardless of error distribution, the OLS estimators become approximately normal, which is exactly what the theory predicts.

Problem 2

a)

The selected dataset contains data for Germany between 1960 and 2023, a description of both variables is presented below.

1. **Households and NPISHs Final Consumption Expenditure (constant 2015 US$):** This variable represents the market value of all goods and services, including durable products (e.g., cars, washing machines), purchased by households and nonprofit institutions serving households (NPISHs). The data is adjusted for inflation, expressed in constant 2015 US dollars. Indicator Code: NE.CON.PRVT.KD. Source: World Bank.
2. **GDP (constant 2015 US$):** This variable represents the gross domestic product, which is the sum of gross value added by all resident producers in the economy plus product taxes and minus subsidies. It is adjusted for inflation and expressed in constant 2015 US dollars. Indicator Code: NY.GDP.MKTP.KD. Source: World Bank.

Both variables have been adjusted to billions of 2015 constant US dollars. The dataset includes 54 observations for each variable, covering the period from 1970 to 2024. Observations with null values were excluded to ensure consistency and reliability in the analysis. Output summary from R:

Variable	Observations	Minimum	Maximum	Mean	Variance
GDP_Billions (X)	54	1411.21	3702.23	2611.74	504318.1
Households_consumption expenditure_Billions (Y)	54	777.77	1905.14	1423.32	111736.7

The data reveal a wide range in GDP, indicating significant economic fluctuations over the observed period. Household consumption expenditure shows a narrower variance, suggesting relatively stable spending patterns. As a substantial share of GDP, household consumption remains consistently high, while the larger volatility in overall GDP may be derivate from other components such as investment, government outlays, or net exports.

b)

R output: Residuals:

Min	1Q	Median	3Q	Max
-88.697	-30.49	-8.061	37.083	68.61

Coefficients:

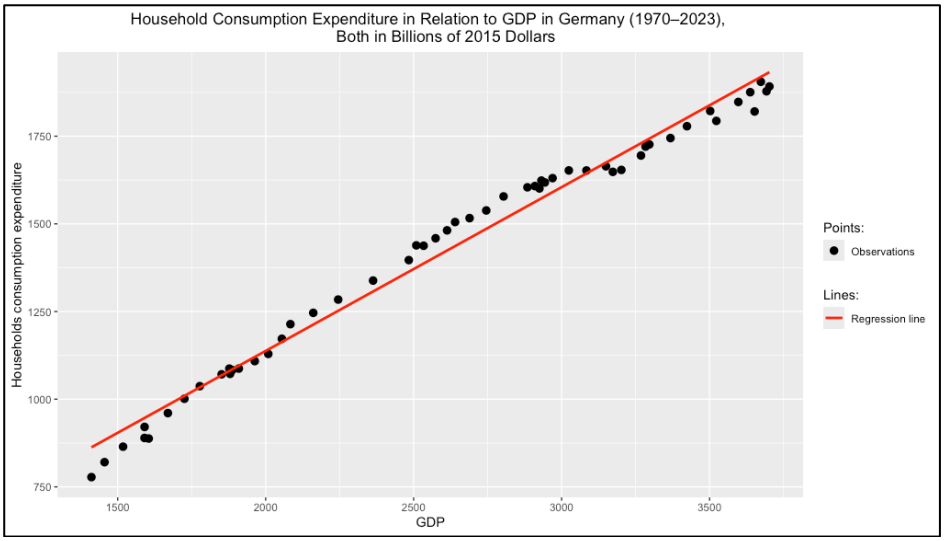
	Estimate	Std.Error	t value	Pr(>|t|)	
(Intercept)	203.7	22.2	9.178	1.84E-12	***
GDP_Billions	0.467	0.00821	56.903	< 2e-16	***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 42.43 on 52 degrees of freedom
Multiple R-squared: 0.9842, Adjusted R-squared: 0.9839
F-statistic: 3238 on 1 and 52 DF, p-value: < 2.2e-16

Fitted model:

Household consumption expenditure= 203.7 + 0.467 GDP

c)



The fitted regression line in red closely follows the data points, indicating a strong linear relationship between GDP and Household Consumption Expenditure.

This is consistent with the model's high R-squared value (0.9842), which suggests that 98.42% of the variation in Household Consumption Expenditure is explained by changes in GDP. The residuals, representing the differences between the observed values and the values predicted by the regression line, are relatively small, with a Residual Standard Error (RSE) of 42.43 billion, and appear to be randomly distributed. This indicates that the model captures the relationship effectively, with no evident systematic bias or pattern in the errors.

d)

The estimated intercept is 203.7 (in billions of 2015 constant US dollars). This represents the theoretical value of Household Consumption Expenditure when GDP is 0.

While this estimate is mathematically valid in the context of the model, it is not realistic or logic in practical terms. A GDP of 0 implies the complete absence of economic activity.

The estimated slope is 0.467, meaning that for every 1 billion increases in GDP, Household Consumption Expenditure is expected to increase by 467 million (in 2015 constant dollars).

This value is logical and consistent with economic theory, as higher GDP typically correlates with higher household consumption due to increased economic activity and disposable income (Mankiw, 2020). The positive slope reflects this relationship and supports the validity of the model within the observed data range.

While the slope provides a meaningful insight into the relationship, the intercept lacks practical interpretability but remains necessary for the regression model's mathematical formulation.

e)

From the regression results: The p-value associated with the slope coefficient for GDP is < 2e-16, which is extremely small. This p-value is smaller than conventional significance level, like α=0.05 or even α=0.001. It indicates strong evidence against the null hypothesis ($H0$:$\beta 1$=0), which means the effect of GDP on Household Consumption Expenditure is statistically significant.

The t-value for the slope is 56.903, which is large. This confirms the statistical significance of the coefficient because it shows that the slope is far from zero in standard error units.

The slope of 0.467 indicates that for every 1 billion increases in GDP, Household Consumption Expenditure is expected to increase by 467 million.

The p-value and t-value provide strong evidence that this relationship is not due to random chance but is a meaningful and statistically significant effect.

f)

The approximate 99% confidence intervals for $\beta 1$(the intercept) and $\beta 2$ (the slope) are as follows:

Coefficient	Lower.Bound..0.005.	Estimate	Upper.Bound..0.995.
Intercept (B_1)	144.3761	203.7248	263.0735
Slope (B_2)	0.4450	0.4670	0.4889

For Intercept β_1, the true value of the intercept lies within this interval [144.38,263.07] billion. Estimate value: 203.72 billion.

For Slope β_2, the true value of the Slope lies within this interval [0.445,0.489]. Estimate value: 0.467

g)

Null Hypothesis (H_0): The slope ($\beta 2$) is less than or equal to 0 ($\beta 2 \leq 0$), this implies that GDP has no positive or significant effect on Household Consumption Expenditure.

Alternative Hypothesis (H_1): The slope ($\beta 2$) is greater than 0 ($\beta 2 > 0$), this implies that GDP positively and significantly affects Household Consumption Expenditure.

Economic theory suggests that household consumption expenditure should increase with GDP, as rising GDP indicates stronger economic activity and higher disposable income. Testing this relationship can confirm the expected positive impact of GDP (Mankiw, 2020).

$t_value: [1] 56.90314 $critical_t:[1] 1.674689 $p_value_test [1] 8.189492e-49.

The calculated t-value is greater than the critical t-value and the p-value is effectively 0, which is below the significance level of 0.05.

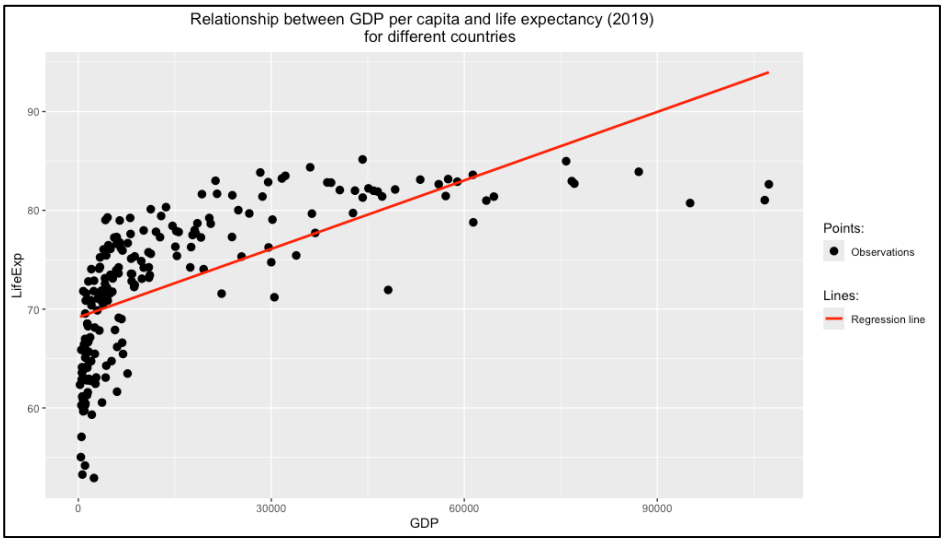
We reject the null hypothesis (H_0) and conclude that there is strong evidence to support the alternative hypothesis (H_1). This means that GDP positively and significantly impacts Household Consumption Expenditure.

Problem 3

a)

The dataset comes from the World Bank World Development Indicators for the year 2019. Variables: **(1) GDP per capita** (Expressed in constant 2015 US$, Indicator Code: NY.GDP.PCAP.KD): GDP (sum of gross value added by resident producers, plus any product taxes and minus subsidies) divided by midyear population. **(2) Life expectancy at birth (total years, Indicator Code: SP.DYN.LE00.IN):** Average number of years a newborn would live if current mortality patterns persisted throughout its life. The dataset includes 200 observations (countries) for the year 2019.

b)

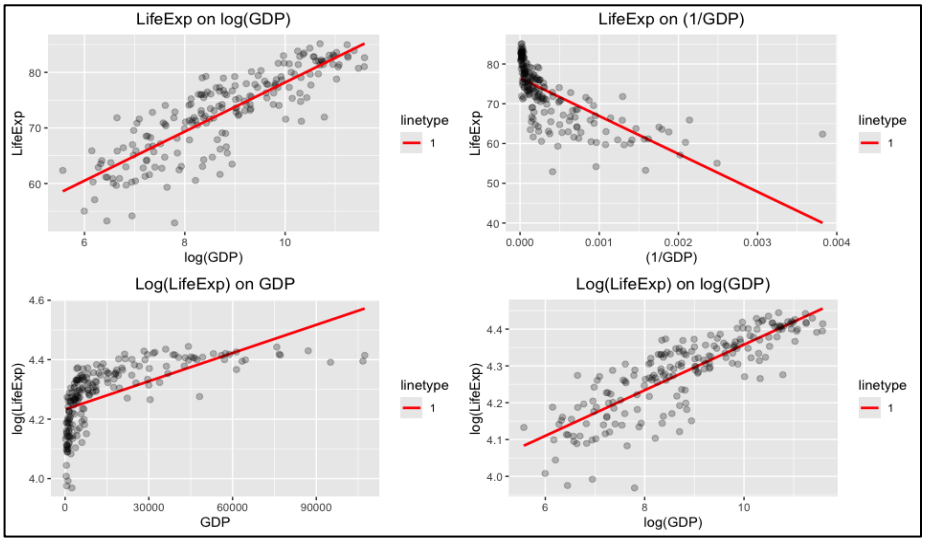


The Figure shows the relationship between GDP per capita and life expectancy for 200 countries in 2019. The red line represents a simple linear regression fit on the data. It is clear that GDP per capita is skewed, it suggests a non-linear relationship: GDP increases sharply at lower life expectancies but

then levels off, implying a diminishing slope at higher life expectancies. A single straight line does not account for this curve.

Applying linear model directly did not capture the true pattern or satisfy the assumptions for a regression. The large residual spread (and potential funnel shape in the scatter) indicates likely heteroscedasticity, the variance of residuals grows with the magnitude of GDP.

c)



Across the four scatterplots, it is evident that in the last graph where X*=log(X) and Y*=log(Y) help to soothe the variability in GDP and life expectancy. Using just X*=Log(X) has a similar result showing a linear pattern between GDP and life expectancy. The inverse transformation X**=1/X appears to be less effective in capturing some kind of linear pattern. Similarly, taking just the logarithm of life expectancy Y*=log(Y) also don't show a linear relationship between variables.

d)

Model	Interc.	Slope	SE_ Interc.	SE_Slope	t-Inter.	t-Slope	p_Intercept	p_Slope	R2
(X, Y)	69.17	2.31×10^{-4}	0.499	1.88×10^{-5}	138.7	12.31	3.61×10^{-199}	3.12×10^{-26}	0.434
logX,Y	33.99	4.42	1.759	0.197	19.3	22.38	1.96×10^{-47}	3.88×10^{-56}	0.717
1/X, Y	76.45	-9.53×10^{3}	0.464	7.02×10^{2}	164.7	-13.57	8.52×10^{-214}	4.20×10^{-30}	0.482
X,Log(Y)	4.23	3.16×10^{-6}	0.007	2.74×10^{-7}	581.2	11.54	5.73×10^{-322}	6.86×10^{-24}	0.402
logX,logY	3.74	6.19×10^{-2}	0.026	2.91×10^{-3}	144.4	21.32	1.25×10^{-202}	3.65×10^{-53}	0.697

1. (X, Y): About 43% of the variance in Y is explained by GDP. The slope is 2.31×10^{-4}, very small in absolute terms but significant ($p<10^{-26}$). GDP is highly skewed, so a straightforward linear model often underperforms.

2. (Log(X), Y) : R^2=0.717: This is the highest explanatory power among the five models, explaining about 72% of the variance in Y. Taking the GDP log significantly improves the fit, perhaps because it linearizes the connection and diminishes the effect of extreme GDP levels. The slope (4.42) shows that for every unit rise in (logX), (Y) increases significantly.

3. ((1/ X), Y): R^2 = 0.482, better than (X, Y) but not as strong as (log X, Y). The negative slope (-9530) is statistically significant but less intuitive economically, and the fit is not as high as with log transformations.

4. (X, log Y): R^2 = 0.402 the lowest among these models. Although logging (Y) alone can help, it appears less effective than logging X (or both).

5. (log X, log Y): R^2 = 0.697 is likewise a good fit. The slope (0.0619) can be interpreted as an elasticity: a 1% rise in GDP is associated with a 0.0619% increase in Y.

e)

Using the slope(β_2) and the intercept(β_1) of the previous question:

For (Xi,Yi*) fitted Model (on the log scale):

$\log(Yi) = \beta_1 + \beta_2 X_i \Rightarrow \hat{Y}_i^* = 4.23 + 3.16\times10^{-6}\ X_i$

Re-transformed Model (back to Y):

$\hat{Y}_i^* \Rightarrow \exp(\hat{Y}_i^*) = \exp(4.23 + 3.16\times10^{-6}\ X_i)$

For (Xi*,Yi*) fitted Model (on the log scale):

$\log(Yi) = \beta_1 + \beta_2 \log(X_i) \Rightarrow \hat{Y}_i^* = 3.74 + 6.19 \times 10^{-2} \hat{X}_i^*$

Re-transformed Model (back to Y):

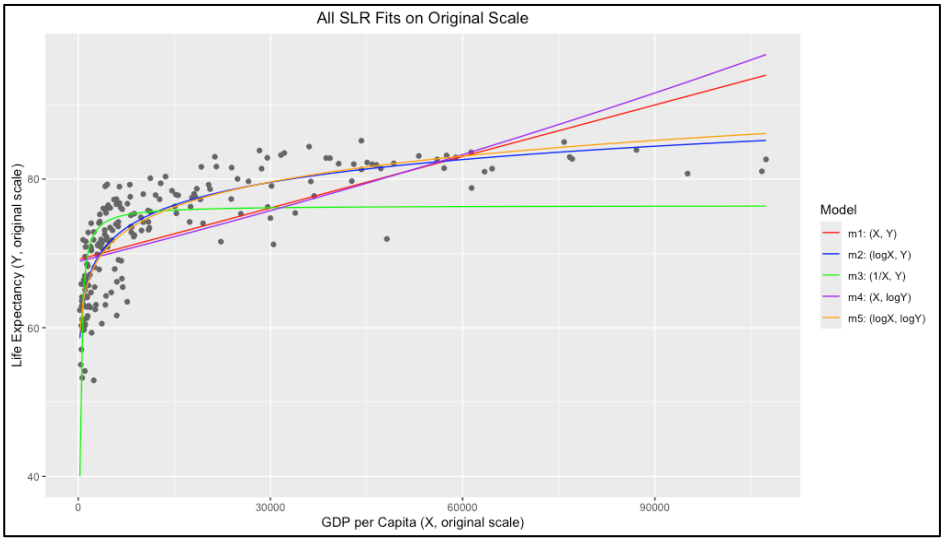
$\hat{Y}_i^* = \exp(\hat{Y}_i^*) = \exp(3.74)\times X_i^{6.19\times10^{-2}}$

f)

Model	RSS
m1 (X, Y)	6296.955
m2 (log X, Y)	3148.747
m3 (1/X, Y)	5757.771
m4 (X, logY)	6630.757
m5 (logX, logY)	3225.215

With the least RSS (3,148.75), Model 2 is better than the others in explaining the relationship between life expectancy and GDP. At 3,225.22, Model 5 performs well as well. By contrast, Models 1, 3, and 4 have larger RSS, suggesting that they don't explain very well the variation in Y. Consequently, Model 2 appears to be the "best" among these options, because it produces the lowest residual sum of squares. Important to note that for models m4 and m5, the dependent variable is on a log scale. So to be comparable, the exponentiate function was used to return it to the **original scale**, comparable with the other models.

g)



The Figure compares the fitted lines from all five models on the original (X,Y) scale. The raw linear model (X,Y) (red line) differs notably from the transformed ones, indicating a less accurate fit in the high-GDP region. Models that use log(X) appear to bend or flatten in ways that better capture the data's curvature and reduce skew. Visually, the(log(X),Y) and (log(X),log(Y)) lines appear to track the main trend more closely across the entire GDP range.

Based on the residual sums of squares (RSS), Model 2 (log(X),Y) has the smallest RSS, suggesting it best fits the data in terms of unexplained variation. Also consider Model 5 (logX,logY) valuable for its elasticity interpretation, though its RSS is slightly higher. Thus, Model 2's smaller RSS implies a better fit statistically.

h)

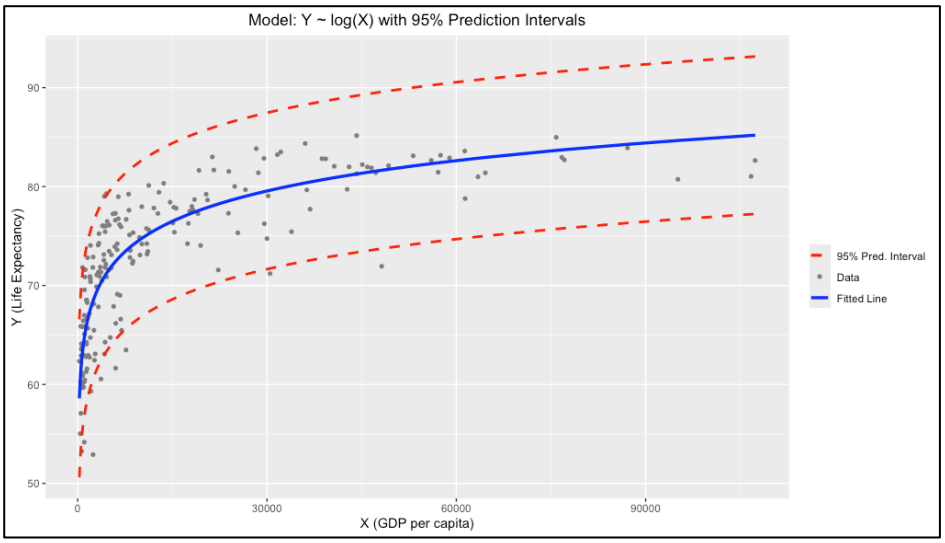


Figure shows the observed data points (X,Y) in gray, alongside the best-fit curve from the model Y ~log(X) in blue, and the 95% prediction intervals in dashed red. The vertical span between the red lines reflects the range where individual future observations (of life expectancy) are likely to fall, given a particular GDP per capita. When life expectancy increases with GDP at a decreasing rate, consistent with a log-transformed relationship. The intervals are relatively tighter at midrange GDP values—where data are more densely clustered—and widen for very high GDP levels, where the model has fewer observations and thus higher uncertainty. Overall, this visualization suggests the logarithmic transformation captures the main pattern well, but predictive accuracy diminishes somewhat at the extremes.

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